The Media Mixer:

User Creativity through Production, Deconstruction and Reconstruction of Digital Media Content

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Abstract

In recent years many museums have experimented with different approaches to involving users through digital media. We explore how remixing and content sharing can be used as a means for user participation. Remix culture is seen as a culture that allows and encourages the production of derivative works; works that are based on already existing works. This cultural practice thrives throughout the Internet, most notably on web2.0 sites like YouTube. The Media Museum has embraced the remix paradigm with the development of an interactive media experience centre called the Media Mixer. Here the museum users can produce, deconstruct, reconstruct and finally publish and share digital media content. The media content is created by the user in the museum's physical environment, but it can be mixed with material from local or global archives. In that way the gap between the analogue and the digital domains is bridged, and the intention is that the users learn about media through a creative process where the border between creating, playing and learning is blurred. There are many challenges in evolving a museum based on analogue print media to an interactive and participatory site but we suggest that the practice of remixing can be a potent means to achieve meaningful user participation in the digital museum age.

The Media Mixer: User Creativity through Production, Deconstruction and Reconstruction of Digital Media

The Media Mixer is a new media experience centre in the Media Museum located in Odense, Denmark. We were both directly involved in developing the Media Mixer: CHM was the overall project manager, and VV designed and implemented one of the exhibits The Media Mixer opened on August 20th 2010 after a long and challenging development process that this paper will elaborate upon. The main focus of the new media centre is the utilization of digital media to promote user participation and the creative production of media content through multimedia editing and "mashups" or "remixes". The Media Mixer features several interactive exhibits revolving around media, such as a chroma key studio, an interactive interview exhibit and a Foley sound booth. All user created content from exhibits is stored digitally and can be edited, shared and published by the users at will.

In our opinion there is a largely unused potential in making mashups of digital content in museums, and we will discuss how and why it was done in the Media Mixer project. Digital mashups are understood as derivative works, and any museum could provide users with content, tools as well as incentives to engage in a process where user generated content and curated content leads to new expressions and understandings.

To include remixing in a museum's practice is simply to reflect what is already happening in society, most prominently on web2.0 sites such as YouTube. Every single minute 24 hours of audiovisual content is uploaded to YouTube (YouTube, n.d.). Of course not all of the content is mashups, but a great deal of the popular content is. Consider for example the remixing of Internet memes such as "The Star Wars Kid" or Bruno Ganz' portrayal of Hitler in the 2004 German film *Der Untergang*. These mashups range from

subtitling a video clip to completely reinventing the material with advanced post-production software, and they are extremely popular. One subtitled parody of *Der Untergang* was viewed on YouTube more than 4 million times before it was removed due to copyright infringement claims (Rohrer, 2010). Copyright issues will be briefly touched upon later. The important point here is that digital remixing is a practice of this generation; a democratic way of dialogue and meaning making. In his book *Remix* (Lessig, 2008) Creative Commons chairman and law professor Lawrence Lessig discusses how the copyright system and the economy need to change to embrace this cultural practice. Lessig draws a distinction between RO (Read-Only) culture and RW (Read/Write) culture where RW culture denotes the practice of remixing where people "add to the culture they read by creating and re-creating the culture around them" (Lessig, 2008 p. 28). This is of course not a new cultural practice, but the tools by which we can create and remix are indeed new. Consider "writing" as an example. A novel could be considered more or less Read-Only. However a scientific article is in some ways Read/Write, because here it is common practice to explicitly draw upon and build on top of other peoples writing. An Internet blog where users can comment, discuss and link is perhaps an even more obvious example of Read/Write expression.

The same democratic forms of creativity and meaning making takes place in multimedia forms of "writing". But the multimedia tools of expression are new and society needs to adapt to the new cultural practices that digital technologies have fostered. Lessig sees an optimistic future with "better RO culture, a more vibrant RW culture, and a flourishing world of hybrids", but as he puts it, this requires "changes in law, and changes in us" (Lessig, 2008 p.252). We suggest that museums could also contribute to a culturally

rich hybrid future where digital remixing is a natural and democratic mode of creative expression and meaning making. But in order to do that, museums too need to change. Not by abandoning centuries of RO museum practice, but by building on top of the existing museum resources and by providing new resources like the Media Mixer.

The Goal of the Media Mixer

We especially wanted to reach and engage a new younger audience (in a 14-24 age bracket). A visit to the Media Mixer should facilitate a more reflective perspective on the information we get from our media as well as inspiration to take a more (inter)active part in the public sphere by engaging in the RW culture of the web. The means to achieve this goal falls into two categories:

- Provide the user with resources they do not have at home in front of their computer.
- Place the user in situations where they can experience the inner workings of media.

The Media Mixer was developed to accommodate single users or small groups rather than entire school classes.

Getting the users actively engaged in creating media content of course aims at offering a certain way of learning about media which other parts of the museum do not facilitate. Since user participation in modern media is a learning point in itself it was natural to make the Media Mixer a place where users are “learning by doing”. We consider digital remixing of media content a useful tool in museums and exhibitions organized on either discovery learning or constructivist lines (Hein, 1998 p. 25): In the remixing process, users construct meaning themselves. The user constructed meaning could be based on knowledge viewed as independent of the learner (which is discovery learning) or knowledge constructed in the

mind of the learner (which is constructivist learning). The Media Mixer contains elements of both discovery learning and constructivist learning. In the interactive exhibits users are able to discover the nuts and bolts of media. For example being interviewed themselves by a virtual interviewer users get a sense of the interview genre and techniques by experiencing it with their own body. This kind of situated learning is supposed to give the user a more profound understanding of how difficult it can be to answer coherently in an interview situation. We view such knowledge as independent of the learner. Remixing the interview with other content afterwards on the other hand does not lead to a discovery of established external knowledge. Instead we view it as an activity where new knowledge is constructed in the mind of the user. So the Media Mixer as a learning site does not subscribe to a certain epistemology but simply aims at letting users construct meaning whether from internal or external knowledge. Other museums would perhaps position themselves in a more radical and fixed epistemological position aiming at either letting users discover established facts or letting users construct their own knowledge. We believe that the different epistemological positions are not necessarily linked to the type of museum, and therefore an art exhibition could be organized on discovery learning just as well as a science exhibition could be organized on constructivist lines.

A detailed view of the Media Mixer

The Media Mixer consists of three studios, a remix worktable, and a small exhibition area. In addition there is also a small cinema and a digital media library where the user can browse through a collection of Danish TV and radio shows. The activities in each of the three studios are dedicated to one specific topic of the media: In *The Hot Seat* the user can engage in an interview with a virtual host portrayed by famous Danish

television reporters. In the *Chroma Key Studio* the user can report "live" from different locations and eras. The user can choose between a positive, negative and neutral speak to be shown on the teleprompter thus emphasising the importance of narrative framing in television reporting. In the *Sound Box* users take on the role of the Foley Artists creating sound effects for film clips by analogue means such as shoes, cardboard boxes etc. hopefully enhancing awareness of the role and inner workings of sound in audiovisual media.

The central hub of the exhibition is the Mixer station, where the users can edit their productions, remixing them with content from the Internet or private sources and finally share their remixes with their friends online as well as with other users in the museum. Because of security issues it was decided not to give the users direct access to the Internet from the Mixer. Instead they have to find the material on two designated internet computers and store it on a USB-key which they can then plug in the Mixer.

For more direct inter-user communication and meaning making the small exhibition area called *The Word is Yours* presents a hot topic from the current Danish media debate. Users are then able to express their own opinion through simple or novel media ranging from Post-it notes, a blackboard, a telephone to a typewriter linked to a digital billboard. The area also features a computer logged on to the Media Mixer weblog serving as a direct link from the physical museum space with the Media Mixer's online domain. This activity is inspired by the Hot Spot methodology with a focus on awareness making on contemporary issues in museums (SAMP 2004). The first dilemma on display was whether erotic manga-style comics and animated films should be considered child pornography and therefore banned like in some other countries. This topic generated major interest from the

media and the museum was the focus of the news cycle for a whole day with appearances on the front page of a daily newspaper, on national radio shows and on primetime TV news. The topic also generated a lot of debate and comments from the users which was of course the most important success criterion.

The concepts for all the activities were initially chosen by museum staff focusing on the points about the workings of our media, that we wanted to convey to the users. At this point the concepts where presented to a group of young test users who rated whether the saw the activities as interesting and gave concrete feedback on the content e.g. which TV presenter the wanted to be interviewed by etc.

Content and copyright

When museum users are encouraged to create, build upon and remix digital content, copyright is of course an issue. This is especially the case when user created content is able to be published directly on the Internet like in the Media Mixer. The copyright laws vary from one country to another so we advice attention toward local law on this area.

Danish law does not have a notion of “fair use”, so in the Media Mixer project, the users are not provided with bits and pieces of commercial content. Instead new content was produced, such as video backgrounds for the Chroma Key Studio and video and sound sequences for the Sound Box. However the most important content is produced by the users themselves and this content is of course theirs to do with what they want. Users must actively click on a file to share it in the museum and online, and the sharing can be undone at any time.

There is of course no way to make sure that users do not bring copyrighted (or offending) material, but should this happen, the museum will remove it when it is discovered. And during the first months of use, it has not happened yet.

Being part of a museum of media history, the Media Mixer features a curated selection of national historical video- and sound clips in the so-called *Mediatheque*. Originally this was hoped to be a resource also to be used in digital user remixes. But this is impossible due to the copyright of the clips. To be even allowed to show the clips in the Media Mixer the museum pays a monthly copyright fee just like a private company would do, and if the content was to be used and published, the fee would be enormous.

Challenges

An unconventional exhibition project such as the Media Mixer faced many other challenges of both an economic, technical and organizational nature. The main economic challenge was that an IT infrastructure with a CMS for handling user profiles with affiliated multimedia content is expensive and do not look like much in an exhibition space: The biggest expenses are almost invisible in the end result.

Regarding technical challenges, what was initially viewed as simple ideas often turned out to be problematic from a developer viewpoint. The museum IT staff consisting of just one system administrator made the museum dependent on the main IT subcontractor to estimate technical challenges and come up with solutions that were realistic technically, economically and time wise. The museum had no option but to trust the professionals until it was too late, and thus core functionalities were not implemented when the exhibition opened.

All in all the greatest technical challenges were related to the handling of user multimedia files – a core feature. From a museum viewpoint the quality of files was too low, audio/video synchronization was inaccurate, audio was missing, videos were upside down, files privileges failed so that one user's video suddenly appeared in another user's files etc.

The organizational challenges were both external and internal. Externally the museum had to draw upon several subcontractors with diverse specialties to realize this project e.g. exhibition architect, programmer, graphic designer, electrician and computer scientist. The museum had to coordinate the actions and different perspectives of these specialists. In our cooperation with each of them the museum was always at a disadvantage because the museum was dealing in areas out of its expertise and unable to properly evaluate if a presented solution was in fact the best. The subcontractors thus had considerable influence on the outcome but no responsibility for how their parts fit together in the greater whole.

Internally a digital project as the Media Mixer had to face entrenched notions of what an exhibition should be and what relation the museum should have with its users. There was a consensus among the curatorial staff at the Media Museum that change and rethinking was needed on these issues as also noted by several museologists (e.g. Hooper-Greenhill 2007 p. 1). But there were different opinions on how radical the change should be. The traditionalists among the staff feared that by letting the users (as co-producers) have considerable influence on what would be shown in the museum space the museum would lose its professional authority and degenerate into a digital playground. The radicalists had no such fears arguing that the users expected to have influence on the content as they are

used to from the Internet and modern RW culture. The solution here was to build the Media Mixer in a separate room with no contact with the museums “real” exhibitions.

The Media Mixer project aims at encouraging users to participate, be creative and reflect upon themselves as both media producers and consumers. This is a big challenge, and therefore the Media Mixer uses different methods for scaffolding the creative processes. We use the term scaffolding to describe all tutoring mechanisms that are aimed at engaging, helping and keeping the users at task (Wood, Bruner & Ross, 1976).

In order to make users explore the different interactive exhibits there are virtually no text in the Media Mixer room. Every interface should speak for itself when users log in with their electronic ticket. When logged in the users get a short range of choices; in scaffolding terms a reduction in the degrees of freedom. In each of the main exhibits users can choose from four to six different predefined tasks. When chosen there is strong direction maintenance in the fact that each task is a narrative (e.g. an interview) taking from around 1 minute to 6 minutes to complete. There are also freestyle modes where users for a short amount of time are able to produce video or audio free of constraints. But tasks that include some predefined video, audio or text material are seen as a necessary means of scaffolding most productions until users are familiar with the production process and inspired to break the boundaries of the simple tasks built into exhibits.

Throughout the exhibits controls are kept as simple as possible in order to focus on the mechanics of media content rather than the mechanics of media production tools. Editing software has been custom made using the same interface style as the exhibits. The museum considered using professional media editing software but chose to mark the critical

features of multimedia editing by providing only simple controls such as cutting and moving blocks of content on just two video tracks and two audio tracks.

The museum hosts are also very active in scaffolding the creative processes. So far they have primarily been active in recruiting, that is encouraging users to try a certain exhibit, as well as in frustration control when exhibits break down. However one of the most powerful scaffolding devices comes from the users themselves. In publishing productions online and onsite users demonstrate different ways of handling the tasks, and hopefully the boundaries of tasks will be crossed and exhibits will be used in creative ways that the museum did not foresee.

Reception

At the time of writing, the Media Mixer has been open to the public for about two months. The reception by both the press and the users has been positive, but the real impact of the Media Mixer experience is somewhat obscured by the fact that there are still technical problems. We have conducted user tests, participant observation and surveys before and after the opening not surprisingly showing that users were enjoying themselves but were frustrated when the software or hardware broke down. Reports of errors or crashes are very consistent among the respondents, and when asked what could be improved in the Media Mixer, the most common answer was “technical stuff”. To our surprise the same respondents generally want to revisit and report that they have really enjoyed themselves. However we cannot rule out the possibility that this is simply a forgiving attitude by users who know that they are experiencing a system that is brand new and therefore not stable yet.

From the first implementation of the system, the museum staff has pointed out interface issues with regard to usability and the user tests have confirmed these issues. One example is the user registration form which is too complicated and takes 2 full minutes to fill out. But the technical problems with basic functionality and robustness have overshadowed almost everything leaving such usability issues more or less untouched.

There are however some consistent findings that are independent of the technical issues. One example is the desire for more content in each exhibit. Users for example want more background videos for the Chroma Key Studio and more video clips in The Sound Box. The reason is not that the users have tried everything: Instead users simply want a large range of choices and are quick to select the one that seems most appealing. Initially we believed that a limited range of choices would serve as a helpful reduction in the degrees of freedom, but the users actually do not need a reduction here. One could speculate that this is simply because the users are used to browse through large quantities of data on the Internet, in their personal music collections and so on. The desire for more content is encouraging for museums wanting to offer access to digital collections, and in the Media Mixer there is a potential for offering media content from the museum's own collections as a production resource for remixing.

Conclusion

The process of evolving a museum based on analogue print media to an interactive and participatory site has been challenging in a lot of ways. Most notably there has been a lot of technical challenges in implementing a system for user multimedia production, remixing and publishing. What the museum initially thought were simple ideas and the developers problems have become the museums and the user's problems.

We however suggest that the practice of remixing can be a potent means to achieve meaningful user participation in the digital museum age. And despite technical issues initial user testing and feedback shows that users are engaged, joyful and willing to produce and share their creative productions.

When working with external developers the key is good communication, realistic project planning with sufficient time allocated for testing and early technical prototyping. We recommend insisting on an iterative process where critical system components are prototyped in the early iterations and demonstrated with placeholder content. Although important, the interfaces and end content should be independent of critical system components, and museums should insist on flexibility in the iterative process where changes in interfaces can be gradually adopted in iterations.

Being a small museum, the Media Museum curators developing content were also the ones highly involved in the technical issues. We suggest that any museum wishing to involve users in a custom made digital exhibition consider the division of assignments so that curators are not too involved in dealing with pure technical problems. The curators' focus should be the tasks, the content and most importantly the users and their creative participation. In the Media Mixer the systems are now beginning to work and therefore the development process which began with the users soon will be all about the users again. All exhibits are developed so that content and tasks can be easily added and modified. User research will show how the complete Media Mixer experience hopefully promotes creativity and reflection works, and how the different tasks should be adjusted to optimize these processes.

References

Hein, G.E. (1998). *Learning in the museum*. London: Routledge

Hooper-Greenhill, E. (2007). *Museums and Education.* London: Routledge

Lessig, L. (2008). Remix: *Making art and commerce thrive in the hybrid economy*. New York: Penguin

Rohrer, F. (April 13 2010) The rise, rise and rise of the Downfall Hitler parody. *BBC News Magazine*. Retrieved fr. http://news.bbc.co.uk/2/hi/uk_news/magazine/8617454.stm Sept. 27th 2010

SAMP (2004) *Hot Spot,*. Sweden: SAMP www.samp.org.

Wood, D.J., Bruner, J.S. & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychiatry and Psychology*. 17 pp. 89-100

YouTube. *YouTube fact sheet*. Retrieved from http://www.youtube.com/t/fact_sheet Sept. 27th 2010